

Project Data Cleaning Report

AI in the Workforce: occupation exposure and task penetration data

STUDENT

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TEAM

PyForce

REPORT DATE

October 4, 2026

SOURCE RELEASE

Anthropic, March 5, 2026

GIT BRANCH

Allen

BRANCH URL

github.com/scha2005/dso576/tree/Allen

PROJECT VISION

Help a career seeker compare career paths by showing how much AI is currently observed across occupations and tasks, while making clear that exposure can reflect assistance or workflow change rather than certain job replacement.

BUSINESS QUESTION

Which occupations have the highest observed AI exposure, and which work tasks show the greatest AI penetration? The primary comparison is `observed_exposure` by O*NET occupation; `penetration` by task is the supporting measure.

Submission metadata: the local branch is `Allen` and tracks `origin/Allen`. The cleaning files are currently uncommitted, so the submitted cleaning commit ID is **pending**. The existing branch head, `6d88b60720007a066dfd92fa61875703b8b2c7bc`, contains only the earlier project-start work and is not presented as the cleaning submission.

What was cleaned

Full-dataset checks

756

job rows retained

17,992

task rows retained

6

exact duplicate rows removed

5 / 5

record checks passed

Original CSVs

Preserved and hashed

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Validate + deduplicate

Stop on malformed data

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Audited outputs

Clean files, samples, evidence

job_exposure.csv

One source row is an occupation. occ_code uniquely identifies all 756 rows. Scores range from 0.0 to 0.7451. There are no missing values, malformed occupation codes, repeated codes, or exact duplicate rows.
Cleaning result: all 756 records retained unchanged.

task_penetration.csv

One cleaned row is a distinct, case-sensitive task description and its penetration score. Scores range from 0.0 to 1.0. Two exact row patterns each appeared four times; one copy of each was retained.
Cleaning result: 17,998 → 17,992 records.

Source and version

Anthropic Economic Index, *Labor market impacts of AI: A new measure and early evidence*, released March 5, 2026; accessed October 4, 2026. The local files were downloaded again at their official revisions and matched byte-for-byte.
job_exposure.csv · revision 7a1cc3c · SHA-256 4f0a3adf5feeb2ec5f5d02ab18cc5e851a2a4b8470bde84c0c9335017be12d68
task_penetration.csv · revision 01eb1d0 · SHA-256 85bee872db1d55d3e9a7f4e89da5ae4a5d59aa8ec875d728fbf4b7d820984616

Interpretation boundary.

Observed exposure is evidence of AI capability and real-world use across work tasks. It is not a probability that a job will disappear, and the supplied task table does not distinguish automation from augmentation on its own.

Rules written before application

Rule · reason · check · result

Rule	Reason	Check	Result
Preserve source files and record SHA-256 hashes.	Keep an auditable input and make reruns comparable.	Hash before processing, hash again afterward, and compare with official file revisions.	PASSED Both unchanged and source-matched.
Require the documented schema.	A missing, extra, or reordered field could invalidate later checks.	Compare CSV headers with the expected columns and stop on malformed rows.	PASSED 0 schema failures.
Keep occupation codes as text matching NN-NNNN.	Identifiers are labels; parsing them as numbers can damage formatting.	Validate all 756 codes with a regular expression and test uniqueness.	PASSED 0 invalid or repeated codes.
Trim outer whitespace only.	Remove accidental padding without rewriting substantive text.	Compare each original field with its stripped value and audit every change.	PASSED 0 fields required trimming.
Parse scores as finite decimals in [0, 1].	The metrics are proportions; invalid or infinite values cannot support comparisons.	Stop on conversion failure or a value outside the allowed interval.	PASSED 0 failures; 0 out of range.
Keep zeros and original numeric precision.	Zero is a valid observed result, not missing data. Rounding could alter rankings.	Count zeros; write the original score strings after validation.	PASSED 411 job and 16,644 task zeros retained.
Keep one copy of each exact row.	The project analyzes distinct task descriptions; repeated identical rows would give those descriptions extra weight.	Count complete-row tuples, audit every removal, and reconcile rows and score sums.	APPLIED 6 task rows removed.
Preserve case and punctuation.	No authoritative identifier shows that similar-looking descriptions are interchangeable.	Keep signatures case-sensitive and inspect the “Web” / “web” pair.	PASSED Both variants retained.
Do not fill missing values; stop instead.	There is no defensible imputation rule for an occupation, task, or exposure score.	Count blank required fields before and after cleaning.	PASSED 0 missing values.
Keep the two tables separate.	The task file has no occupation code or other defensible join key.	Inspect schemas and avoid a text-based or many-to-many merge.	N/A MERGE No merge performed.

Exception and sensitivity note. Deduplicating task descriptions is appropriate for a distinct-description analysis. If the repeated rows encode different O*NET task records whose identifiers were omitted, frequency-weighted analysis should use the original table. The cleaned data therefore must not be treated as an occupation-to-task map.

Inspect and clean the data

Items 1–7

1 State the project vision and question

DONE

Career seekers compare occupations using observed AI exposure and task penetration while avoiding the claim that exposure equals replacement. Business question: which occupations and tasks show the greatest observed AI involvement? Primary comparison: `observed_exposure` by `occ_code`; supporting metric: task penetration. Evidence: `project-start.md` and this report's cover.

2 Define one row

DONE

`job_exposure`: one occupation, uniquely identified by `occ_code` (756/756 unique). `task_penetration`: one cleaned, case-sensitive task description and score, uniquely identified by `task` after six redundant copies are removed. Evidence: `validation.json`.

3 Preserve the source

DONE

Source: Anthropic Economic Index, March 5, 2026 labor-market-impact release; accessed October 4, 2026. Exact revisions and SHA-256 hashes appear in Section 1. Originals remain at project root and are loaded relative to `clean_data.py`. Hashes are verified after processing.

4 Inspect columns and types

DONE

Occupation codes, titles, and task descriptions remain strings. Both score columns were validated by conversion to exact decimal values. Failed conversions: 0 in 756 job rows and 0 in 17,998 task rows. No type repair was needed; scores are written with their source precision.

5 Standardize text

DONE

Rule: trim only outer whitespace. Changed fields: 0. Example before → after: `11-1031` → `11-1031` and `Legislators` → `Legislators`. Meaningful capitalization remains: the otherwise identical task descriptions ending in "Web pages." and "web pages." remain separate.

6 Investigate duplicates

DONE

Jobs: 0 exact duplicate rows and 0 repeated occupation codes. Tasks: 6 excess exact rows across 2 repeated description groups (each occurred 4 times); 1 copy of each group retained. The six removals are listed individually in `cleaning_audit.csv`.

7 Handle missing data

DONE

Before and after counts are 0 for every column: `occ_code`, `title`, `observed_exposure`, `task`, and `penetration`. No records were filled or removed for missingness. The script stops if a required value is blank.

Validate the results

Items 8–14

8 Check suspicious values

DONE

There are no date fields. Scores were checked for failed conversion, non-finite values, and values outside [0, 1]: all counts were 0. Boundary and unusual values were retained as valid: 411 job scores and 16,644 task scores equal 0; one checked task equals 1.0; the maximum occupation score is 0.7451.

9 Verify merges

NOT APPLICABLE

No merge was attempted. `job_exposure.csv` has an occupation code, while `task_penetration.csv` has only task text and score. There is no shared key, so a claimed relationship or unmatched-key count would be misleading.

10 Reconcile changes

DONE

Jobs: 756 → 756; score sum 58.1944 → 58.1944. Tasks: 17,998 → 17,992 because six exact copies were removed; score sum 1201.2546 → 1196.8968. Difference: $4.3578 = 6 \times 0.7263$. No other row-count or score-sum change occurred.

11 Check calculations

DONE

No missing values affect totals, averages, or denominators. All valid zeros remain in calculations. Clean means use $n=756$ occupations and $n=17,992$ distinct task descriptions: 0.0769767 and 0.0665238, respectively. A missing group key would stop the run rather than form a missing group.

12 Verify individual records

DONE

Five expectations were written from source records and cleaning rules, then compared with output: zero boundary, maximum occupation exposure, 1.0 task boundary, and both four-copy capitalization variants. All five matched. Full evidence appears in Section 5 and `record_checks.json`.

13 Test reproducibility

DONE

From the project folder, run `python clean_data.py` with Python 3.10+; no third-party packages are required. A fresh Python 3.14.7 process recreated all seven output files byte-for-byte. Output row counts, unique keys, and all record checks were independently asserted.

14 Document remaining limitations

DONE

Task records lack occupation codes and task IDs; similar wording cannot be safely resolved; deduplication changes occurrence weighting; source metrics do not prove causation or job loss; and salary, growth, location, education, physical-presence, and automation-versus-augmentation variables are absent. These limits constrain career recommendations.

Counts and important totals

Before ↔ after

Dataset / measure	Original	Cleaned	Difference	Explanation
Job rows	756	756	0	No duplicates, missing values, invalid codes, or invalid scores.
Job score sum	58.1944	58.1944	0.0000	All source scores retained.
Job score mean	0.0769767196	0.0769767196	0	Denominator remains 756.
Job score range / zeros	0.0–0.7451 / 411	0.0–0.7451 / 411	0	Valid boundary values retained.
Task rows	17,998	17,992	–6	Two exact rows each had three excess copies.
Task score sum	1201.2546	1196.8968	–4.3578	Six removed copies × 0.7263.
Task score mean	0.0667437826	0.0665238328	–0.0002199498	Exact duplicates no longer receive extra weight.
Task score range / zeros	0.0–1.0 / 16,644	0.0–1.0 / 16,644	0	Removed duplicate scores were nonzero.

Missingness

Every source and cleaned column has zero blanks. Because missing values do not occur, no total or mean requires an adjusted denominator and no imputation uncertainty is introduced.

Numeric validity

Every score converts to a finite decimal in the closed interval [0, 1]. There are zero failed conversions and zero out-of-range values in both datasets.

Text and identifiers

There are zero whitespace changes. All 756 occupation codes match NN–NNNN and remain unique. Capitalization and punctuation remain source-faithful.

Sample selection

Two 25-row samples provide the allowed 50 cleaned rows. They force in score boundaries, the job maximum, and both duplicate groups; remaining rows follow source order.

Metric caution. Score sums appear here only to prove that the cleaning changes reconcile exactly. They are not estimates of total jobs, economic impact, or displacement risk.

Five expected-versus-actual checks

Expectations set before output comparison

1 · Valid zero exposure			MATCH
ORIGINAL 11-1031 Legislators 0.0 Source count: 1	EXPECTED One unchanged row. Zero is valid and must not be treated as missing.	ACTUAL 11-1031 Legislators 0.0 Clean count: 1	
2 · Highest occupation exposure			MATCH
ORIGINAL 15-1251 Computer Programmers 0.7451 Source count: 1	EXPECTED One unchanged row. A high score remains within [0, 1] and is not an error.	ACTUAL 15-1251 Computer Programmers 0.7451 Clean count: 1	
3 · Upper task boundary			MATCH
Task: "Advise customers on use and care of merchandise."			
ORIGINAL Penetration 1.0; source count 1.	EXPECTED One unchanged row. The closed interval allows 1.0.	ACTUAL Penetration 1.0; clean count 1.	
4 · Exact duplicate group, uppercase "Web"			MATCH
Task: "Write interesting and effective press releases, prepare information for media kits, and develop and maintain company internet or intranet Web pages."			
ORIGINAL Penetration 0.7263; four identical rows.	EXPECTED Retain the first exact row; remove three redundant copies.	ACTUAL One row at 0.7263; three audit entries.	
5 · Case-sensitive counterpart, lowercase "web"			MATCH
Task: "Write interesting and effective press releases, prepare information for media kits, and develop and maintain company internet or intranet web pages."			
ORIGINAL Penetration 0.7263; four identical rows.	EXPECTED Retain one lowercase row; keep it distinct from uppercase "Web."	ACTUAL One lowercase row at 0.7263; three audit entries.	

How I checked one Codex suggestion

Independent expectation + reconciliation

SUGGESTION CHECKED

Remove only exact duplicate task rows and preserve the capitalization variants.

Codex identified six redundant rows in `task_penetration.csv`. I did not accept the suggestion based on visual similarity alone. I defined the rule as equality across every CSV field, including case and punctuation.

1 · INSPECT SOURCE

Separately loaded the raw CSV in PowerShell and grouped (`task`, `penetration`) case-sensitively. Exactly two groups occurred four times each; records 17,890–17,897 were then inspected directly.

2 · PREDICT FIRST

Expected 17,992 rows; one uppercase and one lowercase record; six removals; score-sum decrease of $6 \times 0.7263 = 4.3578$.

3 · COMPARE OUTPUT

Checked the clean CSV, six audit entries, validation totals, and two record checks. Every expected value matched.

Conclusion: the suggestion passed the defined check. It removes six exact copies while keeping the “Web” and “web” descriptions as two separate records. The remaining risk is semantic: the source lacks task IDs, so duplicate-looking rows could represent distinct upstream task records. That uncertainty is documented rather than hidden.

07 · REPRODUCIBILITY

Recreate the submitted results

1. Place the unchanged `job_exposure.csv` and `task_penetration.csv` next to `clean_data.py`.
2. Use Python 3.10 or newer. The script uses only the standard library.
3. From the project folder, run `python clean_data.py`.
4. Review `cleaned/validation.json`, `cleaned/cleaning_audit.csv`, and `cleaned/record_checks.json`.

Restart test: a new Python 3.14.7 process regenerated seven output files. Every file was byte-identical to the prior run; output row counts and key uniqueness were asserted; all five record checks passed.

Expected clean outputs

- `job_exposure_clean.csv`
- `task_penetration_clean.csv`
- two 25-row sample CSVs
- audit CSV and two JSON evidence files

Dependency statement

No external packages are required to clean or validate the data. PDF rendering is separate from the cleaning pipeline and does not affect submitted CSV results.

What the cleaned data cannot establish

Documented, unresolved constraints

No occupation-to-task link

The task table has no occupation code or task ID. It cannot be joined safely to the occupation table, and it cannot show which listed tasks belong to a specific career.

Duplicate identity is uncertain

Identical visible rows may be redundant copies or distinct upstream O*NET records whose identifiers were omitted. The clean file supports distinct-description analysis; the source supports occurrence-weighted sensitivity checks.

Exposure is not replacement

Observed exposure combines capability and use. It can represent automation, augmentation, or workflow change and is not a forecast that an occupation will disappear.

Career criteria are incomplete

Salary, employment growth, location, education costs, worker preferences, physical presence, soft skills, and regional demand are absent. Career recommendations require auxiliary sources.

Platform and time scope

The measures reflect the release’s methodology and observed Claude usage. Adoption patterns, models, and deployment contexts may change, so later releases may not be directly comparable without version controls.

Similar text is unresolved

The rules do not collapse paraphrases, spelling variants, or capitalization differences. Doing so would require an authoritative task identifier or a documented semantic-matching method.

Evidence map

File	Purpose
clean_data.py · cleaning_notebook.ipynb · requirements.txt	Cleaning script, restartable review notebook, and standard-library dependency statement.
CLEANING_README.md	Source/version, rules, run instructions, output list, and limitations.
cleaned/*_clean.csv	Full cleaned job and task tables.
cleaned/*_sample.csv	Two 25-row samples, including difficult cases.
cleaned/cleaning_audit.csv	Six row-level duplicate removals with source record numbers.
cleaned/validation.json	Counts, missingness, duplicate evidence, ranges, totals, means, and hashes.
cleaned/record_checks.json	Five original/expected/actual comparisons and pass results.

References

Anthropic. “Labor market impacts of AI: A new measure and early evidence.” March 5, 2026. anthropic.com/research/labor-market-impacts

Anthropic Economic Index, labor_market_impacts data directory. huggingface.co/datasets/Anthropic/EconomicIndex

PyForce project vision and candidate assessment: project-start.md.

Before final submission: commit and push the cleaning deliverables to branch Allen, then replace the pending commit note on the cover with the submitted commit ID and regenerate this PDF.